NexusPay

Smart Digital Payment Solution

SOFTWARE PROJECT REPORT

NexusPay

Smart Digital Payment Solution



Introduction

NexusPay is a secure and modern digital payment platform designed to simplify financial transactions. It enables users to perform instant money transfers, QR payments, bill payments, transaction tracking, and digital wallet management through an intuitive mobile application while maintaining high security and reliability.

Business Problem

Traditional payment systems often suffer from slow processing, limited accessibility, security concerns, and poor user experience. NexusPay addresses these challenges by providing a fast, secure, scalable, and user-friendly digital payment ecosystem that improves transaction efficiency and enhances customer convenience.

Core Modules

- User Authentication & Security
- Digital Wallet Management
- Money Transfer
- QR Code Payments
- Bill Payments
- Transaction History
- Notifications & Alerts
- Profile Management
- Admin Dashboard
- Analytics & Reporting

Secure & Reliable
Bank-level security to protect your money

Fast Transactions
Instant payments in just a few seconds

Accessible Anywhere
Access your wallet anytime, anywhere

Smart Analytics
Track spending and manage finances better

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CONNECT • TRANSACT • GROW

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OBJECTIVES

Driving Efficiency. Reducing Risk.
Creating Value.

01



Objective 1:
Improve Payment Processing Efficiency

- Achieve a transaction success rate of at least 95%.
- Reduce failed transactions by 20%.
- Process over 100,000 transactions with accurate record management.



02



Objective 2:
Strengthen Fraud Detection and Risk Management

- Identify 100% of high-value transactions above ₹50,000 for review.
- Reduce fraudulent transaction losses by 15%.
- Monitor suspicious user activities through automated SQL queries.



03



Objective 3:
Generate Actionable Business Insights

- Identify the top 10% of users contributing to transaction volume.
- Measure monthly transaction growth and user activity trends.
- Increase customer retention by 10% through reward and cashback analysis.



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ER DIAGRAM
NEXUSPAY SYSTEM

Structured and normalized data model for efficient transaction processing, analytics and risk management.

DIMENSIONS

Bank_Accounts
- account_id INT
- user_id INT (FK)
- bank_name VARCHAR(100)
- account_number VARCHAR(50)
- ifsc_code VARCHAR(20)
- account_type VARCHAR(20)
- branch VARCHAR(100)
- is_primary BOOLEAN

PK account_id

Wallets
- wallet_id INT
- user_id INT (FK)
- balance DECIMAL(18,2)
- wallet_status VARCHAR(20)
- wallet_type ENUM(...)
- created_at DATETIME

PK wallet_id

Merchants
- merchant_id INT
- merchant_name VARCHAR(100)
- merchant_category VARCHAR(50)
- city VARCHAR(50)
- contact_email VARCHAR(100)
- created_at DATETIME

PK merchant_id

NBFC_Partners
- nbfc_id INT
- name VARCHAR(100)
- license_no VARCHAR(50)
- status ENUM(...)
- contact_person VARCHAR(100)
- email VARCHAR(100)
- phone BIGINT
- commission_rate DECIMAL(5,2)
- max_daily_loan DECIMAL(18,2)

PK nbfc_id

User_Info
- user_id INT
- user_fullname VARCHAR(100)
- user_username VARCHAR(50)
- user_email VARCHAR(100)
- user_phone BIGINT
- user_dob DATE
- user_city VARCHAR(100)
- user_registration_date DATE
- max_status VARCHAR(20)

PK user_id

KYC
- kyc_id INT
- user_id INT (FK)
- document_type ENUM(...)
- document_number VARCHAR(100)
- verification_status ENUM(...)
- verification_date DATETIME
- expiry_date DATE

PK kyc_id

Fraud_Alerts
- alert_id INT
- transaction_id INT (FK)
- risk_level VARCHAR(20)
- fraud_reason VARCHAR(200)
- alert_date DATETIME
- is_resolved BOOLEAN

PK alert_id

Transactions (FACT TABLE)
- transaction_id INT
- user_id INT (FK)
- bank_account_id INT (FK)
- wallet_id INT (FK)
- sender_account_id INT (FK)
- receiver_account_id INT (FK)
- nbfc_id INT (FK, Nullable)
- amount DECIMAL(18,2)
- transaction_type VARCHAR(20)
- payment_method VARCHAR(20)
- transaction_date DATETIME
- transaction_status VARCHAR(20)

USER_INFO (Inside Transaction)
- user_id INT
- user_fullname VARCHAR(100)
- user_username VARCHAR(50)
- user_email VARCHAR(100)
- user_phone BIGINT
- user_dob DATE
- user_city VARCHAR(100)
- user_registration_date DATE
- user_status VARCHAR(20)

PK transaction_id

Rewards
- reward_id INT
- user_id INT (FK)
- transaction_id INT (FK)
- reward_points INT
- points_value DECIMAL(18,2)
- reward_date DATE
- expiry_date DATE
- is_redeemed BOOLEAN

PK reward_id

Customer_Satisfaction
- csat_id INT
- transaction_id INT (FK)
- rating INT
- nps_score INT
- review_comments TEXT
- category ENUM(...)
- feedback_channel ENUM(...)
- created_at DATETIME

PK csat_id

Loans
- loan_id INT
- user_id INT (FK)
- nbfc_id INT (FK)
- partner_id INT (FK)
- loan_amount DECIMAL(18,2)
- interest_rate DECIMAL(5,2)
- tenure_months INT
- loan_status ENUM(...)
- disbursed_date DATE
- closure_date DATE (Nullable)

PK loan_id

Insurance_Policies
- policy_id INT
- user_id INT (FK)
- partner_id INT (FK)
- policy_type ENUM(...)
- policy_amount DECIMAL(18,2)
- premium_amount DECIMAL(18,2)
- policy_status ENUM(...)
- start_date DATE
- end_date DATE
- claim_amount DECIMAL(18,2)

PK policy_id

EMI_Payments
- emi_id INT
- loan_id INT (FK)
- user_id INT (FK)
- emi_amount DECIMAL(18,2)
- payment_date DATE
- due_date DATE
- payment_status ENUM(...)
- payment_mode ENUM(...)

PK emi_id

RELATIONSHIP LEGEND
- One-to-Many (Dimension to Fact)
- Many-to-One (Fact to Dimension)

KEY LEGEND

PK	Primary Key
FK	Foreign Key
ENUM	Enumeration
DATETIME	Date & Time
DECIMAL	Numeric Value

DIMENSIONS
Master data that provides context to the facts.

FACT TABLES
Contains measurable business events and transactional data.

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BASIC SQL QUERIES

Essential Queries for Data Exploration and Daily Operations



01 Retrieve All Users

Get all user information.

```
SELECT *
FROM user_info;
```

INSIGHT
Helps view complete user database for overview and checks.

02 Users from a City

Get users from a specific city.

```
SELECT user_id, full_name, city, mobile_number
FROM user_info
WHERE city = 'Delhi';
```

INSIGHT
Useful for location-based analysis and targeted offers.

03 Active Users

Get all active users.

```
SELECT user_id, full_name, mobile_number, account_status
FROM user_info
WHERE account_status = 'Active';
```

INSIGHT
Helps in monitoring active users base for business tracking.

04 Bank Accounts by User

Get bank accounts of a user.

```
SELECT *
FROM bank_accounts
WHERE user_id = 1;
```

INSIGHT
Useful to check linked accounts for a specific user.

05 Wallet Balance by User

Get wallet balance of a user.

```
SELECT wallet_id, wallet_balance, wallet_status
FROM wallets
WHERE user_id = 1;
```

INSIGHT
Provides quick view of wallet balance and status.

06 All Merchants

Get all merchant information.

```
SELECT *
FROM merchants;
```

INSIGHT
Helps explore all merchant data for merchant analytics.

07 Transactions by User

Get all transactions of a user.

```
SELECT *
FROM transactions
WHERE user_id = 1
ORDER BY transaction_date DESC;
```

INSIGHT
Useful to view user transaction history recent to old.

08 Successful Transactions

Get successful transactions.

```
SELECT *
FROM transactions
WHERE transaction_status = 'Success';
```

INSIGHT
Helps in identifying successful payment trends.

09 Failed Transactions

Get failed transactions.

```
SELECT *
FROM transactions
WHERE transaction_status = 'Failed';
```

INSIGHT
Helps identify failed payments for issue resolution.

10 Recent Transactions

Get recent transactions.

```
SELECT *
FROM transactions
ORDER BY transactions_date DESC
LIMIT 10;
```

INSIGHT
Gives quick overview of the latest platform activity.

KEY BENEFITS

- Faster Reporting
- Fraud Detection
- Business Intelligence
- Performance Monitoring
- Customer Analytics

TABLES COVERED

- user_info
- transactions
- insurance_policies
- fraud_alerts
- bank_accounts
- loans
- recharge
- customer_satisfaction
- wallets
- emi_payments
- rewards
- kyc
- merchants

USE CASES

- Payment Analytics
- Fraud Detection
- Customer Insights
- User Monitoring
- Financial Reporting
- Business Intelligence

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02

ADVANCED SQL QUERIES
PART 1 (1–5)

Business Intelligence Queries for Transaction Analytics and Decision Support



01 Total Transactions Per Day

Get total number of transactions and amount for each day.

```
SELECT DATE(transaction_date) AS Transaction_Date,
       COUNT(*) AS Total_Transactions,
       SUM(amount) AS Total_Amount
FROM transactions
GROUP BY DATE(transaction_date)
ORDER BY Transaction_Date;
```

INSIGHT

Helps in monitoring daily transaction volume and revenue trends.

02 Top 5 Customers by Transaction Amount

Find top 5 customers based on total transaction amount.

```
SELECT u.user_id, u.full_name,
       SUM(t.amount) AS Total_Spent
FROM users u
JOIN transactions t ON u.user_id = t.sender_user_id
GROUP BY u.user_id, u.full_name
ORDER BY Total_Spent DESC
LIMIT 5;
```

INSIGHT

Identifies high-value customers contributing maximum to business.

03 Successful vs Failed Transactions

Count successful and failed transactions.

```
SELECT transaction_status,
       COUNT(*) AS Total_Count
FROM transactions
GROUP BY transaction_status;
```

INSIGHT

Gives an overview of transaction success rate for operational efficiency.

04 Merchant Revenue Report

Get total transactions and revenue generated by each merchant.

```
SELECT m.merchant_id, m.merchant_name,
       COUNT(t.transaction_id) AS Total_Transactions,
       SUM(t.amount) AS Revenue
FROM merchants m
JOIN transactions t ON m.merchant_id = t.merchant_id
GROUP BY m.merchant_id, m.merchant_name
ORDER BY Revenue DESC;
```

INSIGHT

Helps evaluate merchant performance and revenue contribution.

05 Wallet Balance Report

Get current wallet balance and status of all users.

```
SELECT u.user_id, u.full_name, w.wallet_balance,
       w.wallet_status
FROM wallets w
JOIN users u ON w.user_id = u.user_id
ORDER BY w.wallet_balance DESC;
```

INSIGHT

Provides real-time view of wallet balances and user wallet status.

KEY BENEFITS

- Faster Reporting
- Fraud Detection
- Business Intelligence
- Performance Monitoring
- Customer Analytics

TABLES COVERED

- user_info
- transactions
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- fraud_alerts
- bank_accounts
- loans
- recharge
- customer_satisfaction
- wallets
- emi_payments
- rewards
- kyc
- merchants

USE CASES

- Payment Analytics
- Fraud Detection
- Customer Insights
- User Monitoring
- Financial Reporting
- Business Intelligence

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03

ADVANCED SQL QUERIES

PART 2 (6–10)

Business Intelligence Queries for Transaction Analytics and Decision Support



06 Transaction Summary

Get total transactions and amount for each month.

```
SELECT DATE_FORMAT(transaction_date, '%Y-%m') AS Month,
       COUNT(*) AS Total_Transactions,
       SUM(amount) AS Total_Amount
FROM transactions
GROUP BY DATE_FORMAT(transaction_date, '%Y-%m')
ORDER BY Month;
```

INSIGHT

Helps in tracking monthly growth and business performance trends.

07 Average Transaction Amount

Get the average amount of each transaction.

```
SELECT AVG(amount) AS Average_Transaction_Amount
FROM transactions;
```

INSIGHT

Helps understand average spending behavior of customers.

08 Top Recharge Service Providers

Find top 5 recharge service providers by revenue.

```
SELECT service_provider,
       COUNT(*) AS Total_Recharges,
       SUM(amount) AS Total_Revenue
FROM recharge
GROUP BY service_provider
ORDER BY Total_Revenue DESC
LIMIT 5;
```

INSIGHT

Shows revenue by recharge services and revenue contribution.

09 Active Loan Summary

Get summary of all active loans.

```
SELECT l.loan_id, l.loan_amount, l.interest_rate,
       l.loan_status
FROM loans l
WHERE l.loan_status = 'Active'
ORDER BY l.loan_id;
```

INSIGHT

Helps monitor active loans and customer loan portfolio.

10 EMI Payment Status Report

Get EMI payment status summary.

```
SELECT payment_status,
       COUNT(*) AS Total_EMI_Payments,
       SUM(amount) AS Total_Amount
FROM emi_payments
GROUP BY payment_status
ORDER BY Total_Amount DESC;
```

INSIGHT

Helps track EMI collections and pending payments for follow-up.

KEY BENEFITS

- Faster Reporting
- Fraud Detection
- Business Intelligence
- Performance Monitoring
- Customer Analytics

TABLES COVERED

- user_info
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- insurance_policies
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USE CASES

- Payment Analytics
- Fraud Detection
- Customer Insights
- User Monitoring
- Financial Reporting
- Business Intelligence
- Business Intelligence

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04

ADVANCED SQL QUERIES

PART 3 (11–15)

Business Intelligence Queries for Transaction Analytics and Decision Support



11 Peak Transaction Hours

Find the busiest hours of the day.

```
SELECT HOUR(transaction_date) AS Hour,
       COUNT(*) AS Total_Transactions,
       SUM(amount) AS Revenue
FROM transactions
GROUP BY HOUR(transaction_date)
ORDER BY Total_Transactions DESC;
```

INSIGHT

Identifies peak business hours for better server scaling and marketing scheduling.

12 Dormant Users

Find users who haven't made a transaction in the last 90 days.

```
SELECT u.user_id, u.full_name,
       MAX(t.transaction_date) AS Last_Transaction
FROM users u
LEFT JOIN transactions t
ON u.user_id = t.sender_user_id
GROUP BY u.user_id, u.full_name
HAVING MAX(t.transaction_date) <
  DATE_SUB(CURDATE(), INTERVAL 90 DAY);
```

INSIGHT

Useful for identifying inactive users and running re-engagement campaigns.

13 Wallets with Low Balance

Identify users with low wallet balance.

```
SELECT u.full_name, w.wallet_balance
FROM wallets w
JOIN users u
ON w.user_id = u.user_id
WHERE w.wallet_balance < 500
ORDER BY w.wallet_balance;
```

INSIGHT

Helps target low-balance users with cashback, top-up and promotional offers.

14 Fraud Risk Analysis

Find transactions flagged as high risk.

```
SELECT transaction_id, risk_level, fraud_reason, alert_date
FROM fraud_alerts
WHERE risk_level = 'High'
ORDER BY alert_date DESC;
```

INSIGHT

Enables quick identification and investigation of high-risk fraud transactions.

15 Top Reward Earners

Find users with the highest reward points.

```
SELECT u.full_name,
       SUM(r.reward_points) AS Total_Points
FROM rewards r
JOIN users u ON r.user_id = u.user_id
GROUP BY u.full_name
ORDER BY Total_Points DESC
LIMIT 10;
```

INSIGHT

Identifies top reward earners and loyal customers for premium reward programs.

KEY BENEFITS

- Faster Reporting
- Fraud Detection
- Business Intelligence
- Performance Monitoring
- Customer Analytics

TABLES COVERED

- users
- transactions
- wallets
- fraud_alerts
- rewards
- merchants
- emi_payments
- loans
- insurance_policies
- bank_accounts
- recharge
- customer_satisfaction
- kyc
- nbfc_partners

USE CASES

- Payment Analytics
- Fraud Detection
- Customer Insights
- User Monitoring
- Financial Reporting
- Business Intelligence

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SWOT ANALYSIS - NexusPay



Evaluating our Strengths, Addressing Weaknesses, Seizing Opportunities, and Mitigating Threats

STRENGTHS

- Integrated platform offering payments, loans, insurance, and rewards in a single ecosystem.
- Real-time transaction processing and digital wallet convenience.
- Data-driven insights through centralized customer and financial data.
- Fraud monitoring and risk management using advanced analytics.
- User-friendly interface with smooth and secure experience.
- Strong partnerships with banks, NBFCs, and service providers.

WEAKNESSES

- Dependence on third-party NBFCs, banks, and payment gateways.
- Potential system downtime and transaction failures.
- Customer trust can be affected by security incidents or service delays.
- High operational and compliance costs.
- Limited offline usability in low-connectivity areas.
- Complex integration with multiple service partners.

OPPORTUNITIES

- Growing digital payments market and cashless economy.
- Expansion into Tier 2/3 cities and rural markets.
- Introduction of new financial products like mutual funds and savings.
- Partnerships with more merchants and service providers.
- Leveraging AI/ML for personalized offers, fraud prevention, and credit scoring.
- Government initiatives promoting digital finance and inclusion.

THREATS

- Intense competition from fintech and big tech players.
- Cybersecurity threats, data breaches, and fraud attempts.
- Regulatory changes and compliance requirements.
- Economic fluctuations affecting customer spending and loan repayments.
- Technology disruptions and emerging alternatives.
- Negative customer experience spreading through reviews and media.

KEY STRENGTHS THAT DRIVE US

- Secure & Reliable Platform
- Fast & Seamless Transactions
- Data-Driven Decision Making
- Advanced Fraud Protection
- Strategic Partnerships

FOCUS AREAS

- Strengthen Infrastructure
- Enhance Customer Trust
- Improve Operational Efficiency
- Expand Market Reach
- Invest in Innovation

OUR STRATEGY

- Build Trust
- Drive Growth
- Stay Compliant
- Deliver Value
- Innovate Continuously

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ROOT CAUSE ANALYSIS

NEXUSPAY

This analysis identifies the underlying reasons for operational inefficiencies and customer dissatisfaction within the NexusPay ecosystem. Delays in customer support and limited user awareness affect overall customer experience. Inefficient business processes, including loan approvals and complaint resolution, can lead to service delays and reduced customer trust.

ROOT CAUSE CATEGORIES

People:
Customer support delays,
limited user awareness

Process:
Slow loan approvals,
inefficient complaint handling

Technology:
Payment failures,
system performance issues

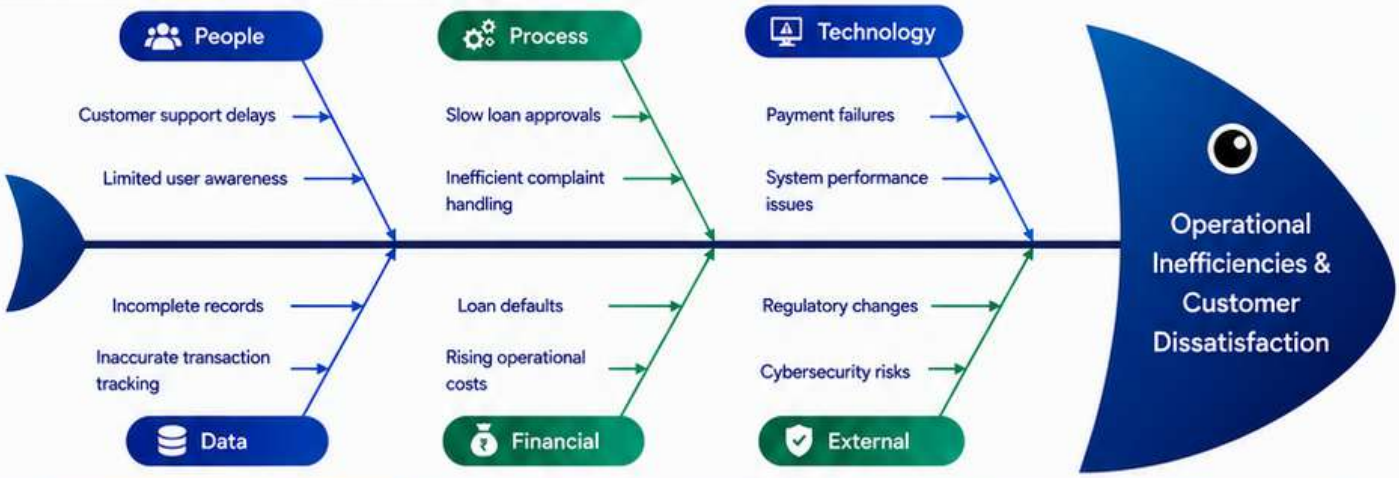
Data:
Incomplete records,
inaccurate transaction tracking

Financial:
Loan defaults,
rising operational costs

External:
Regulatory changes,
cybersecurity risks

This analysis highlights the key areas requiring improvement to enhance operational efficiency, strengthen risk management, and improve customer satisfaction across the NexusPay platform.

ROOT CAUSE (FISHBONE) DIAGRAM



KEY STRENGTHS THAT DRIVE US

- Secure & Reliable Platform
- Fast & Seamless Transactions
- Data-Driven Decision Making
- Advanced Fraud Protection
- Strategic Partnerships

FOCUS AREAS

- Strengthen Infrastructure
- Enhance Customer Trust
- Improve Operational Efficiency
- Expand Market Reach
- Invest in Innovation

OUR STRATEGY

- Build Trust
- Drive Growth
- Stay Compliant
- Deliver Value
- Innovate Continuously

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IMPLICATION REPORT



The implementation of the NexusPay Relational Database Management System (RDBMS) provides a centralized and structured platform for managing digital payments, loans, insurance policies, fraud monitoring, and customer feedback. The system enhances data accuracy, reduces redundancy, and improves operational efficiency through effective data integration and management.

By enabling real-time tracking of transactions, EMI payments, fraud alerts, and customer satisfaction metrics, the database supports informed decision-making and strengthens risk management capabilities. The structured data model also facilitates faster reporting, improved regulatory compliance, and enhanced customer service.

Furthermore, the system creates opportunities for advanced analytics, allowing the organization to identify customer behavior patterns, monitor financial performance, optimize lending operations, and personalize services. Overall, the NexusPay RDBMS implementation contributes to better business performance, increased customer trust, and sustainable growth within the fintech ecosystem.

OPERATIONAL IMPACT

- Centralized data management
- Improved operational efficiency
- Faster reporting and analysis
- Reduced manual errors
- Streamlined business processes

STRATEGIC IMPACT

- Better decision-making
- Enhanced risk management
- Improved regulatory compliance
- Stronger data governance
- Scalable and future-ready platform

CUSTOMER IMPACT

- Enhanced customer experience
- Faster issue resolution
- Personalized services
- Increased customer trust
- Higher satisfaction and loyalty